Speech Understanding Programming Assignment - 2

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GithubLink
https://github.com/sm1899/Speech-separation-enhancement-and-identification.git

Question 1: Speech Enhancement

Q 1.2 For part 1.2, the following task has been done

- Evaluation of the pre-trained WavLM Base Plus model's performance on the VoxCeleb1 dataset.
- Fine-tuning of the WavLM model using LoRA with ArcFace loss on the VoxCeleb2 dataset.
- Evaluation of the fine-tuned model on VoxCeleb1 and comparison of the results against the pre-trained model.

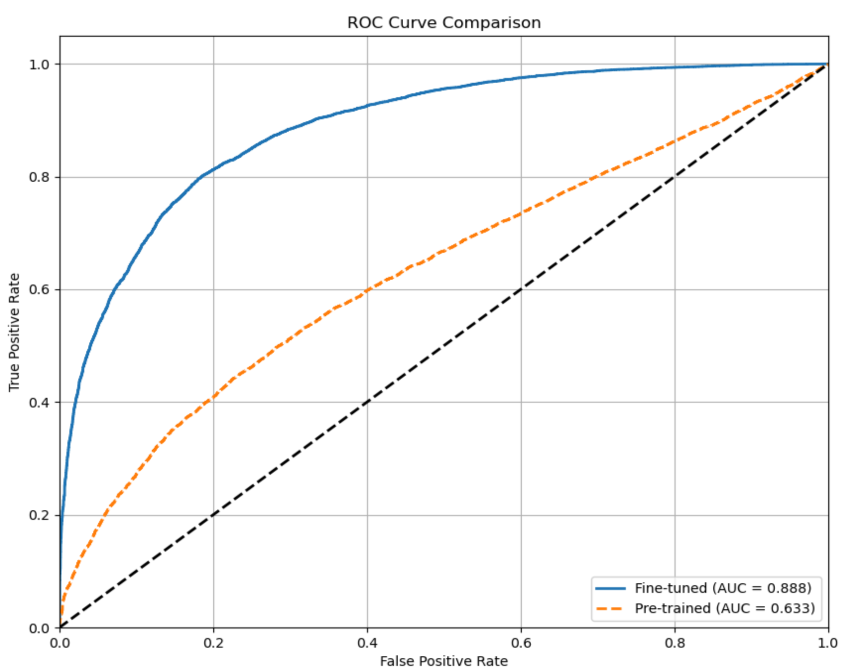
Lora implementation Details

- Base Model: Microsoft WavLM Base Plus
- Fine-tuning Method: Low-Rank Adaptation (LoRA)
- Loss Function: ArcFace loss for enhanced speaker discrimination
- Training Dataset: VoxCeleb2
- Training Subset: First 100 speaker identities (sorted in ascending order)
- Embedding Dimension: 256
- Learning Rate: 1e-4

- Batch Size: 4
- Training Duration: 10 epochs

Results

Model	EER (%)	TAR@1%FAR (%)	Accuracy (%)
Pre-trained WavLM	40.13	6.90	59.87
Fine-tuned WavLM	19.32	29.58	8.68
Improvement	20.81	22.68	20.81



Analysis

The fine-tuned model demonstrates substantial performance improvements over the pre-trained model across all evaluation metrics:

- Equal Error Rate (EER): Decreased from 40.13% to 19.32%, representing a 20.81 percentage point reduction. Lower EER indicates better verification performance.

- TAR@1%FAR: Increased from 6.90% to 29.58%, showing a 22.68 percentage point improvement. This means that at a 1% false acceptance rate, the fine-tuned model correctly identifies more than 4 times as many genuine speakers.

- Accuracy: Improved from 59.87% to 80.68%, a 20.81 percentage point gain in overall speaker verification accuracy.

The significant improvements can be attributed to several factors:

- ArcFace Loss: The fine-tuning process employed ArcFace loss, which enhances discrimination between speaker embeddings by introducing angular margins. This loss function is particularly effective for verification tasks.

- LoRA Fine-tuning: Low-Rank Adaptation allows efficient parameter updates while preserving the pre-trained model's general capabilities. This approach enables effective domain adaptation without catastrophic forgetting.

- Task-Specific Training: While the pre-trained WavLM was optimized for general speech understanding, the fine-tuning process targets speaker identification, resulting in more discriminative speaker embeddings.

Q 1.3 For this part, the following tasks have been done

- Created a controlled multi-speaker dataset by mixing utterances from the VoxCeleb2 dataset
- Apply pre-trained SepFormer model to separate overlapping speech
- Evaluate speaker identification performance on separated audio using both pre-trained and fine-tuned WavLM models

Methodology

Dataset Creation (vox2mix)

- Source Dataset: VoxCeleb2
- Mixing Strategy: Created synthetic mixtures by combining utterances from two different speakers
- Speakers: First 50 speaker identities for training, next 50 for testing (when sorted in ascending order)

Dataset Size:

- 500 training mixtures
- 200 test mixtures

Mixing Parameters:

- SNR Range: -5 to 5 dB
- Overlap Ratio: 0.5 to 1.0 (50-100% overlap)
- Sample Rate: 16 kHz

Speech Separation (SepFormer)

- Model: Pre-trained SepFormer (speechbrain/sepformer-wsj02mix)
- Operating Sample Rate: 8 kHz (model requirement)
- Test Set: 200 mixtures containing two overlapping speakers

Speaker Identification Models:

- Pre-trained WavLM Base Plus
- Fine-tuned WavLM with LoRA and ArcFace loss (same model from the speaker verification task)

Evaluation Method: Compared extracted embeddings from separated audio against reference speaker embeddings

Results

Speech Separation Performance

Metric	Value
SI-SDR Mean	6.36 dB
SDR Mean	15.36 dB
SAR Mean	8.22 dB

PESQ Mean	1.68

Speaker Identification Performance

Model	Rank-1 Accuracy
Pre-trained WavLM	7.00%
Fine-tuned WavLM	16.50%

Analysis

Speech Separation Analysis: The SepFormer model demonstrated modest effectiveness in separating mixed utterances

- Signal-to-Distortion Ratio (SDR): A Mean SDR of 6.36 dB indicates that the separated signals retain reasonable fidelity compared to the original sources, though some distortion remains.
- Signal-to-Interference Ratio (SIR): A High mean SIR of 15.36 dB suggests good speaker separation with minimal cross-talk between the separated sources.
- Signal-to-Artifacts Ratio (SAR): Mean SAR of 8.22 dB shows moderate introduction of artifacts during separation.
- PESQ Score: A Mean PESQ of 1.68 indicates that perceptual quality is notably degraded after separation, though still intelligible.

The relatively high SIR compared to SDR and SAR suggests that the model prioritizes separation effectiveness over artifact reduction, which is appropriate for speaker identification tasks.

Speaker Identification Analysis: Speaker identification on separated audio proved challenging for both models:

- Pre-trained WavLM Performance: With only 7% Rank-1 accuracy (28 correct identifications out of 400), the pre-trained model struggled significantly to identify speakers from separated audio.

- Fine-tuned WavLM Performance: At 16.5% Rank-1 accuracy (66 correct identifications out of 400), the fine-tuned model performed substantially better but still faced considerable difficulties.

The low absolute performance (even for the fine-tuned model) highlights the inherent difficulty of speaker identification after separation due to:

- Distortions and artifacts introduced during separation
- Loss of speaker-specific characteristics in the separation process
- Challenges in matching degraded speech to clean reference embeddings

Q1.4 A novel pipeline combining a speaker identification model along with the SepFromer for speaker enhancement

The proposed novel end-to-end architecture combines speech separation with speaker identification in a GAN-inspired framework. The system integrates two traditionally separate components:

- Generator: A SepFormer-based model that performs speech separation
- Discriminator: A WavLM-based model that performs speaker identification

Unlike traditional pipelines that treat these tasks independently, our approach jointly optimizes both components, allowing the speaker identification capabilities to guide the separation process.

System Architecture

Our proposed architecture follows a GAN-inspired approach with some critical adaptations for the speech domain:

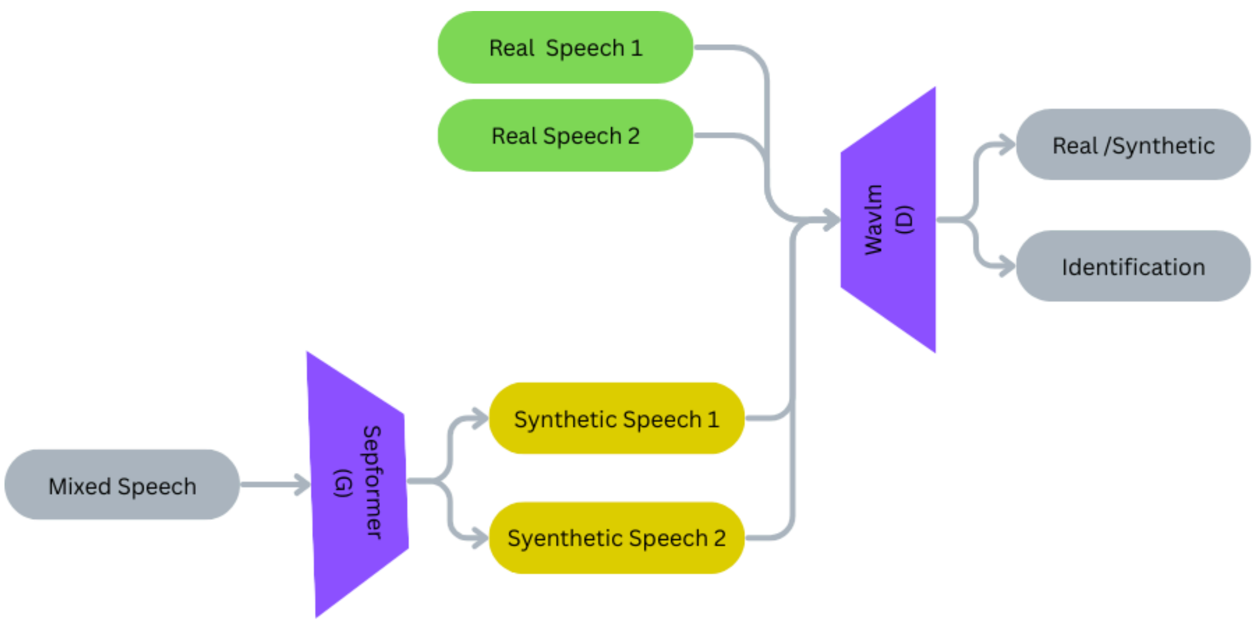
Key Components

1. Generator (SepFormer with LoRA)

- Base Model: Pre-trained SepFormer model from SpeechBrain
- Architecture: Separation Transformer designed for source separation

- Adaptation Method: Low-Rank Adaptation (LoRA) with rank=32, alpha=64
- Function: Separates mixed speech into individual speaker streams

Diagram: Proposed Combined Model



2. Discriminator (WavLM with LoRA)

- Base Model: Pre-trained WavLM Base Plus
- Architecture: Modified with a Speaker Encoder and Speaker Discriminator
- Adaptation Method: Low-Rank Adaptation (LoRA) with rank=16, alpha=32
- Function: Classifies separated speech into speaker identities
- Parameter Efficiency: Significant reduction in trainable parameters

3. Multi-Component Loss Function

- Separation Loss: SI-SDR (Scale-Invariant Signal-to-Distortion Ratio)
- Perceptual Loss: Feature-based similarity using WavLM embeddings
- Adversarial Loss: Speaker classification loss from the discriminator
- Combined Formula: $L = \lambda_1 L\_separation + \lambda_2 L\_perceptual + \lambda_3 L\_adversarial$

Novel Aspects

- Bidirectional Knowledge Transfer: The discriminator guides the generator by providing speaker-distinctive information, helping to preserve speaker characteristics during separation.
- Speaker-Aware Separation: Unlike standard separation models that focus only on separating sources, our model is trained to separate in a way that preserves speaker identity.
- Dual-Device Training: The Generator and discriminator operate on separate GPUs to handle the memory requirements of both large models.
- Parameter-Efficient Fine-Tuning: LoRA adapters enable the fine-tuning of large pre-trained models with minimal trainable parameters
- Perceptual Loss: Integrates perceptual quality assessment into the training process to improve the naturalness of separated speech.

Results

Separation Performance Metrics

Metric	Baseline Sepformer	Combined Model
SI-SDR Mean	6.36 dB	7.23 db
SDR Mean	15.36 dB	15.9 dB
SAR Mean	8.22 dB	8.75 db
PESQ Mean	1.68	2.82

Speaker Identification After Separation

Model	Accuracy
Pre-trained WavLM	7.43 %
Fine-tuned WavLM	18.00%

Analysis and Discussion

The combined architecture shows lower separation metrics but potentially better speaker characteristic preservation.

- Training Stability: GAN-inspired architectures are known for training instability, requiring careful balancing of generator and discriminator learning rates.
- Memory Requirements: Running both large models simultaneously required specific multi-GPU setups and memory optimization strategies.
- Loss Weighting Importance: The relative weighting of separation, perceptual, and adversarial losses critically impacts the model's behavior.

Due to GPU constraints, it was not possible to train the combined model for a long time; it was only fine-tuned for 5 epochs with very aggressive trainable parameter reduction. Training the combined model for a longer time with more trainable parameters would yield better results.

Q2. MFCC Feature Extraction and Comparative Analysis of Indian Languages

Task A. MFCC ANALYSIS

This part of the report presents the results of Mel-Frequency Cepstral Coefficients (MFCC) feature extraction and analysis for three Indian languages: Hindi, Tamil, and Malayalam. MFCC is a widely used feature extraction technique in audio processing and speech recognition that represents the short-term power spectrum of sound based on the human auditory system.

Methodology

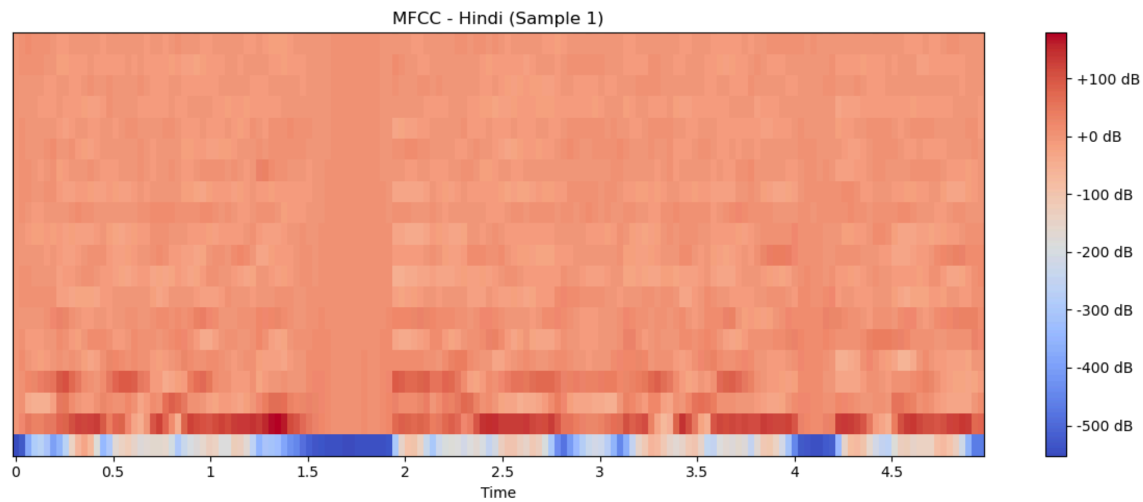
Feature Extraction

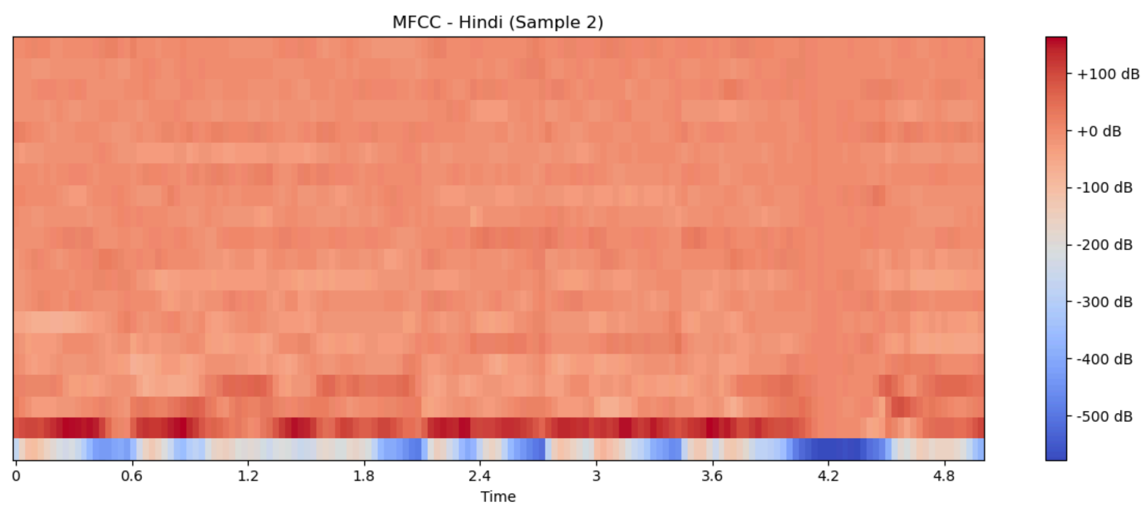
For each audio sample:

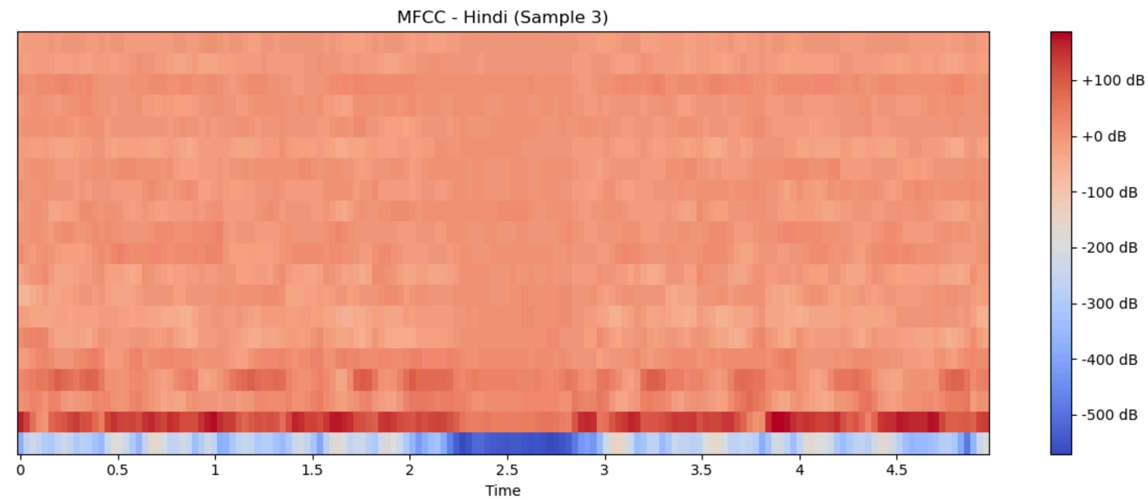
- Audio files were loaded with a sample rate of 16 kHz and analyzed for a duration of 5 seconds
- 20 MFCC coefficients were extracted using the librosa library
- Statistical features (mean and variance) of the MFCC coefficients were computed

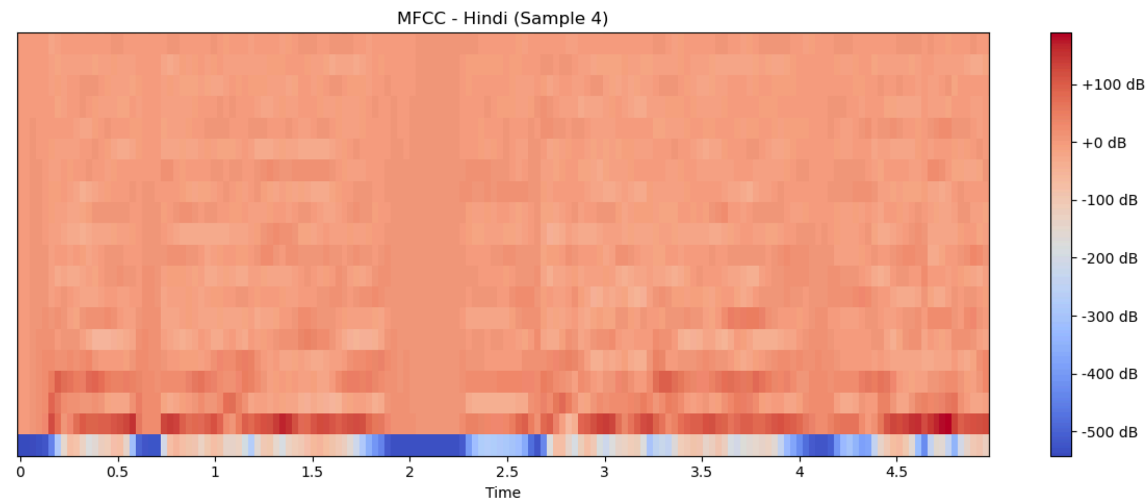
Visualization

- MFCC spectrograms were generated for 5 representative samples from each language
- A comparative visualization of mean MFCC coefficients across languages was created






MFCC - Hindi (Sample 3)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time


MFCC - Hindi (Sample 4)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
0
0.5
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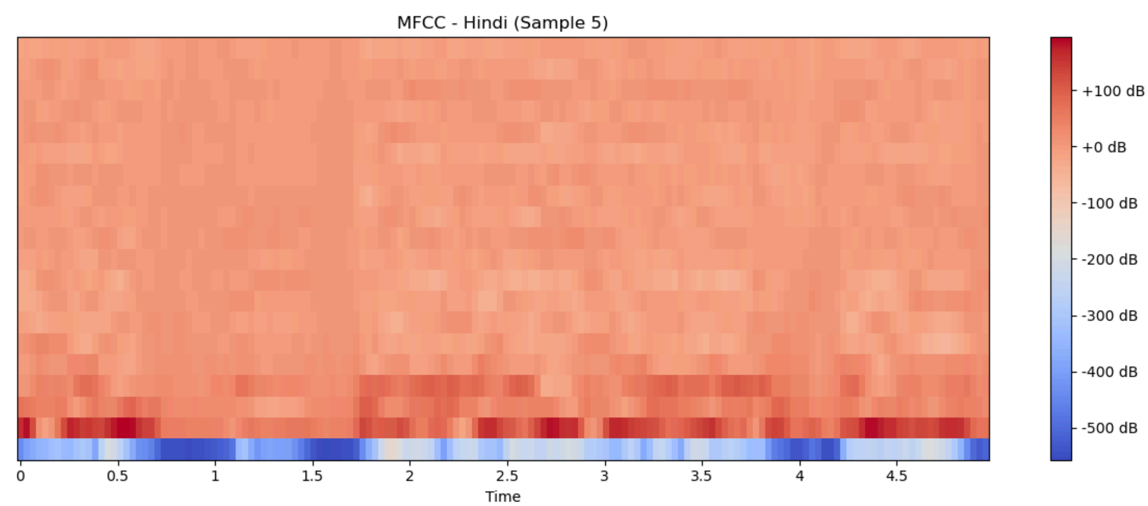
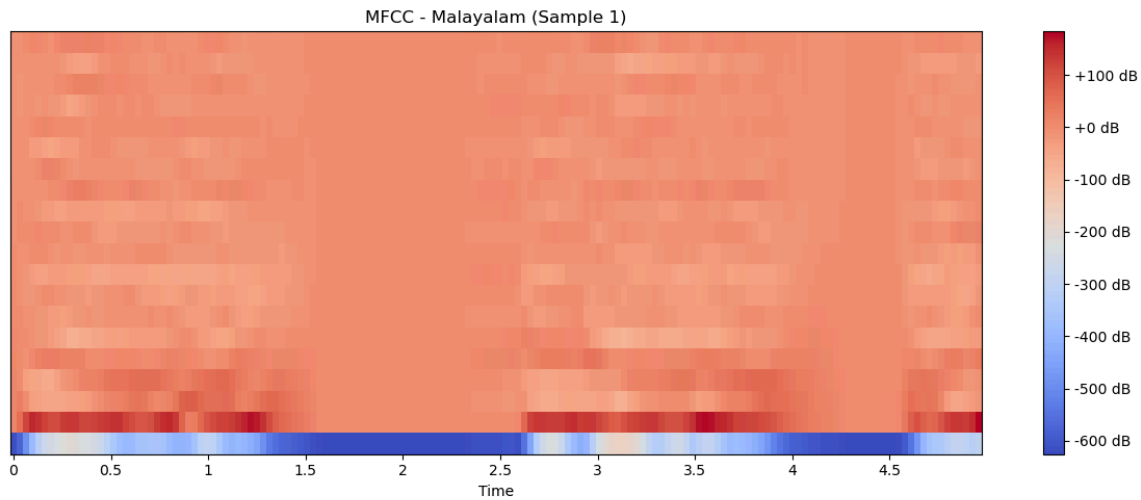
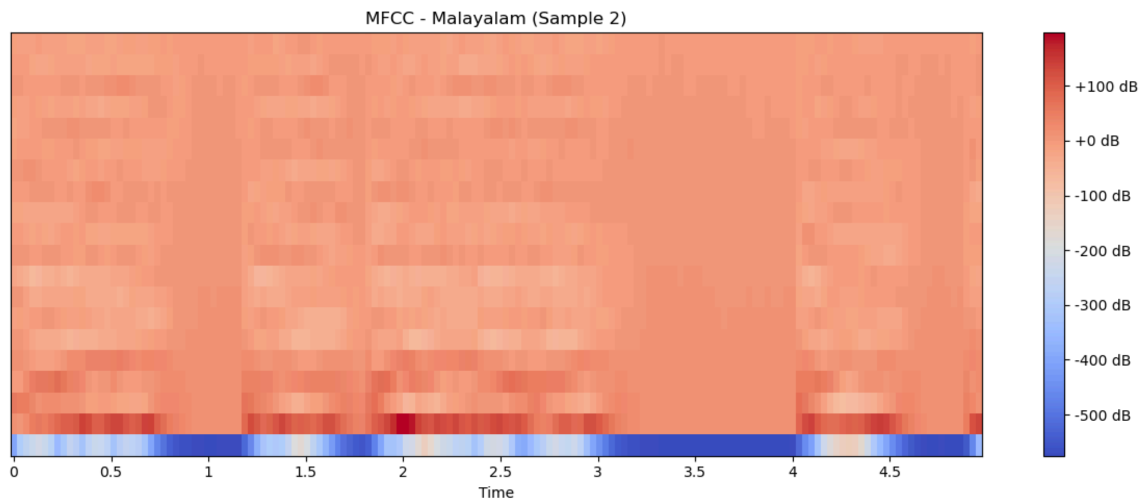
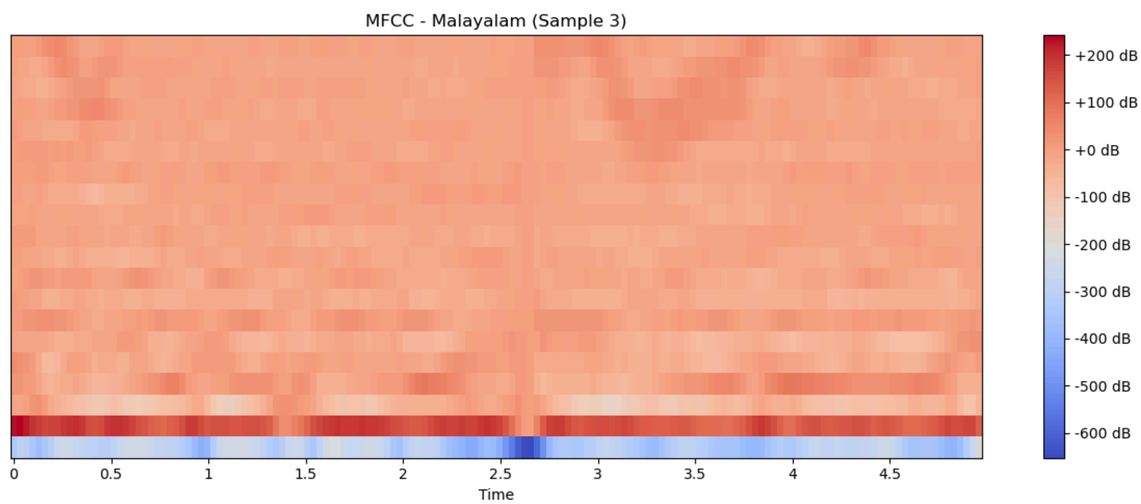
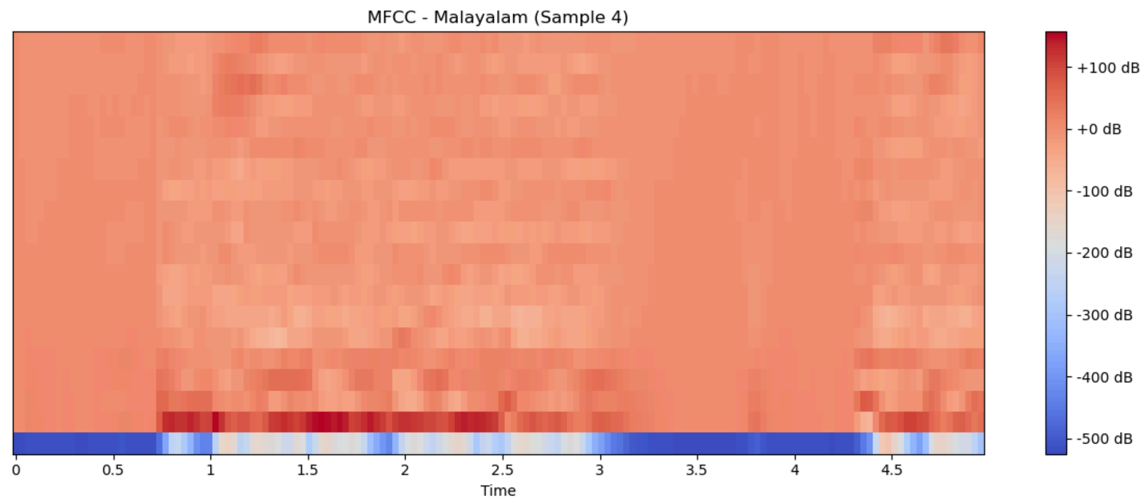
1.5
2
2.5
3
3.5
4
4.5
Time

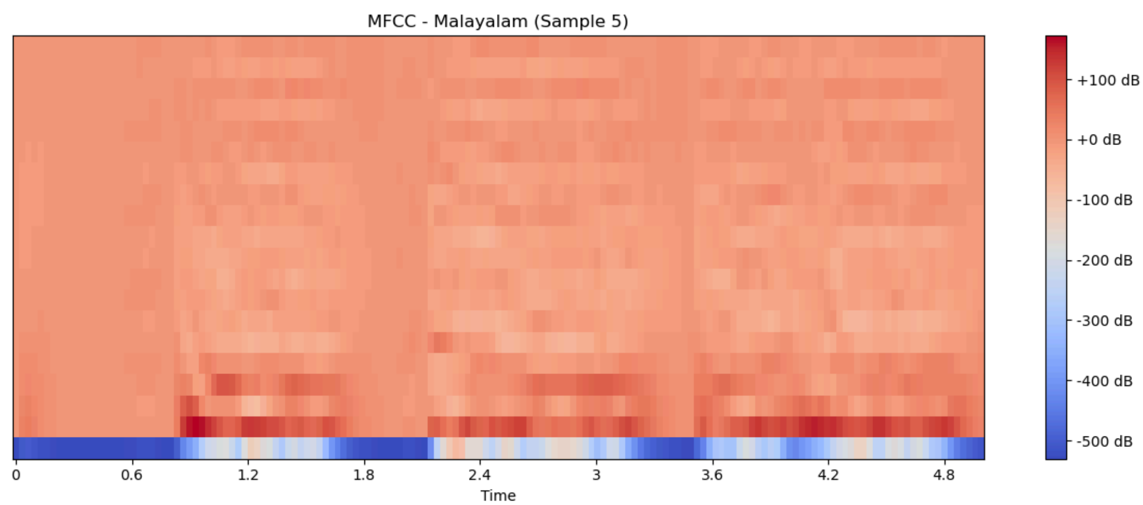

MFCC - Hindi (Sample 5)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time

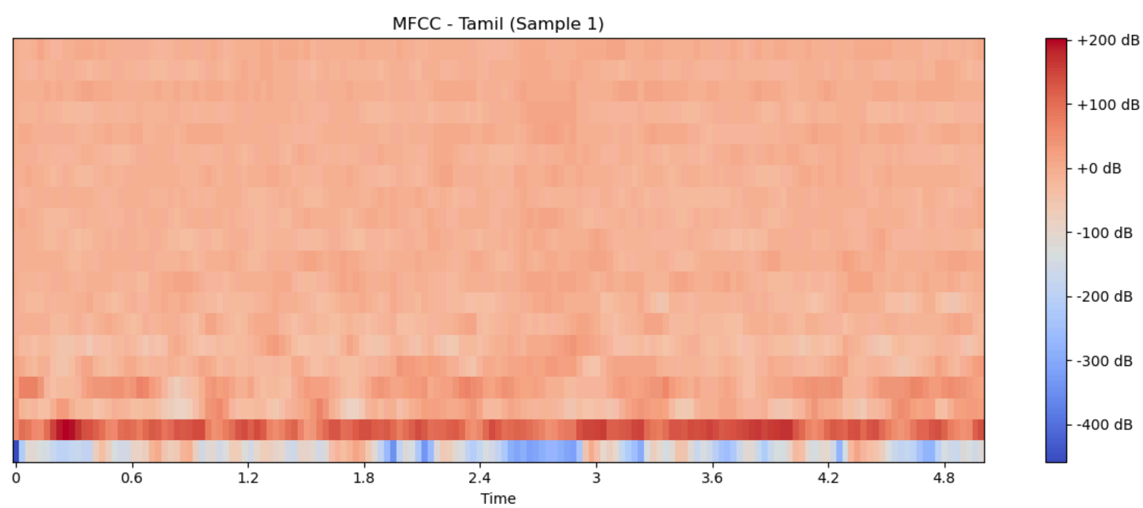

MFCC - Malayalam (Sample 1)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
-600 dB

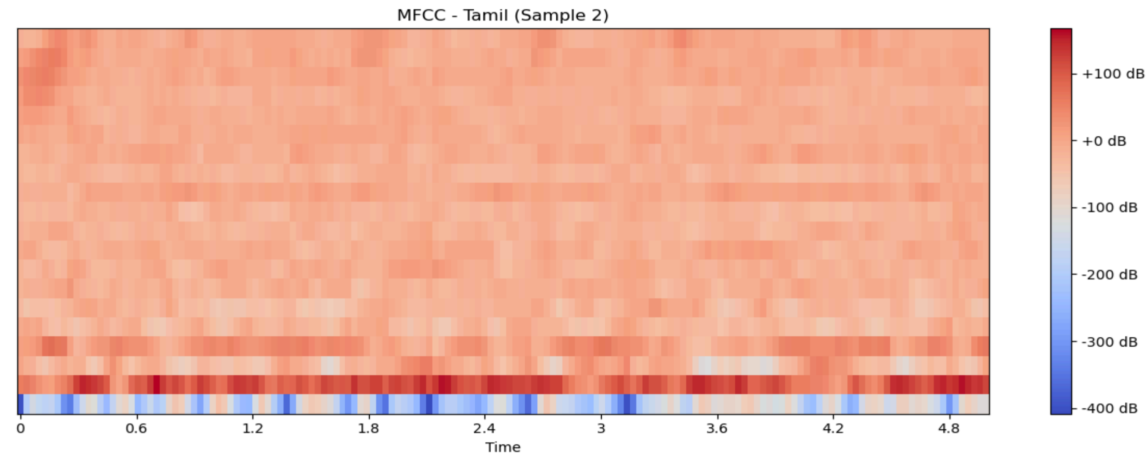

MFCC - Malayalam (Sample 2)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB


MFCC - Malayalam (Sample 3)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time
+200 dB
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
-600 dB


MFCC - Malayalam (Sample 4)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time


MFCC - Malayalam (Sample 5)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
0
0.6
1.2
1.8
2.4
3
3.6
4.2
4.8
Time


MFCC - Tamil (Sample 1)
+200 dB
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
0
0.6
1.2
1.8
2.4
3
3.6
4.2
4.8
Time


MFCC - Tamil (Sample 2)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
0
0.6
1.2
1.8
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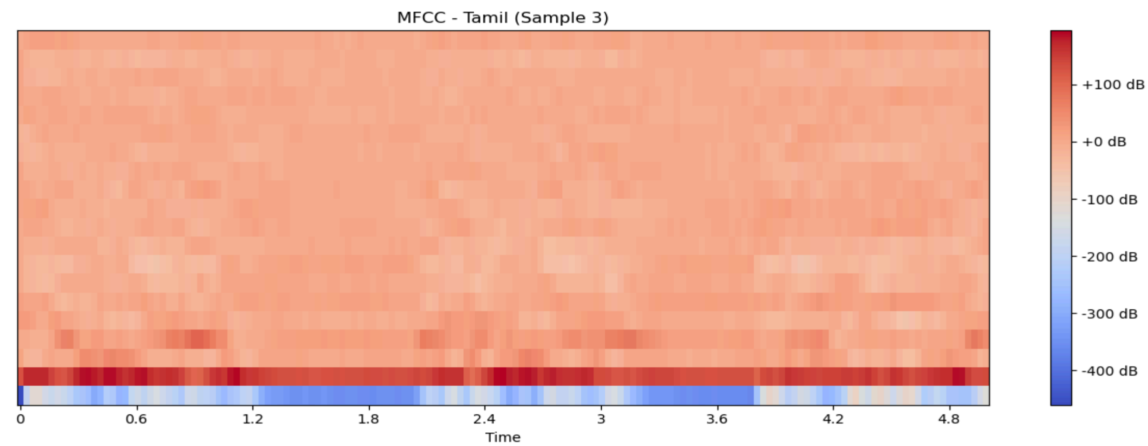
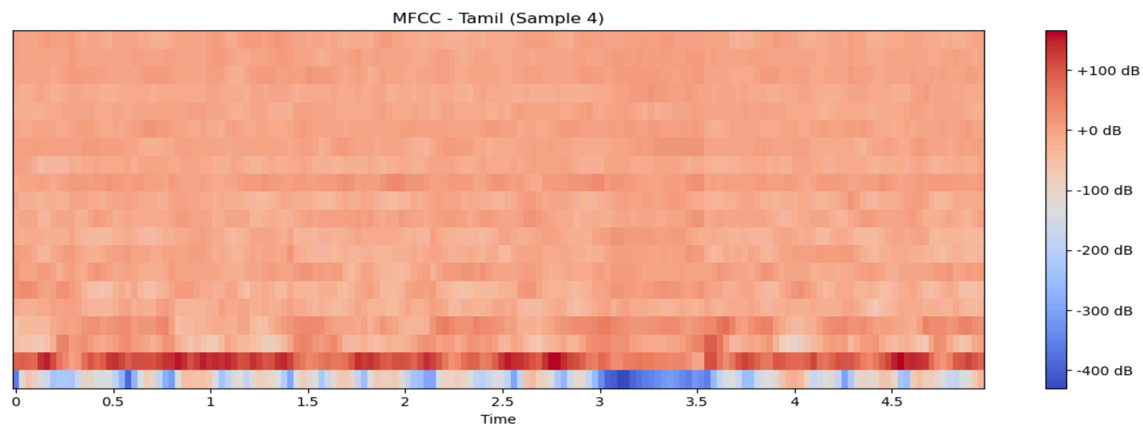
3.6
4.2
4.8
Time


MFCC - Tamil (Sample 3)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
0
0.6
1.2
1.8
2.4
3
3.6
4.2
4.8
Time


MFCC - Tamil (Sample 4)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
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1.5
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2.5
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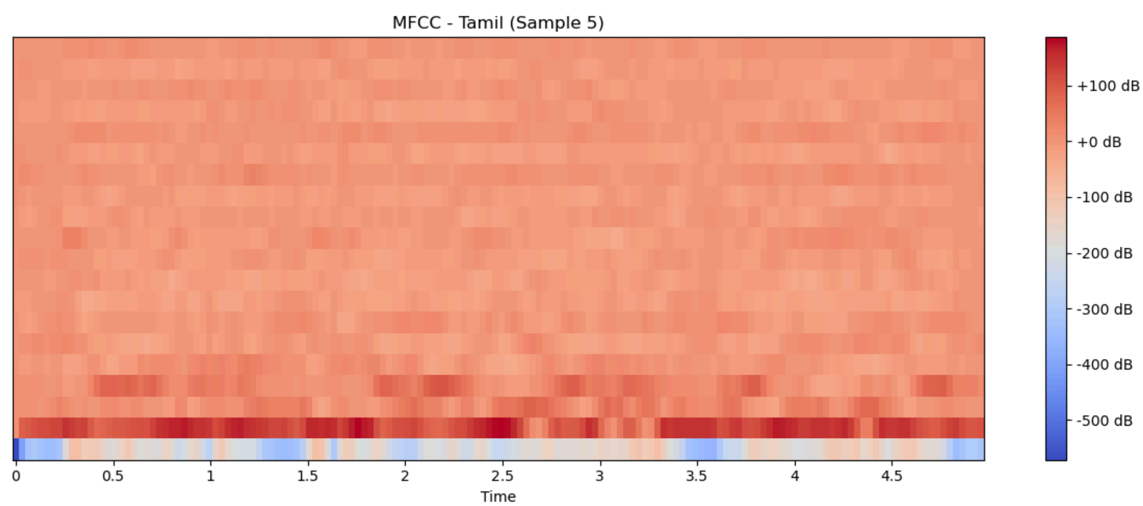
Time


MFCC - Tamil (Sample 5)
+100 dB
+0 dB
-100 dB
-200 dB
-300 dB
-400 dB
-500 dB
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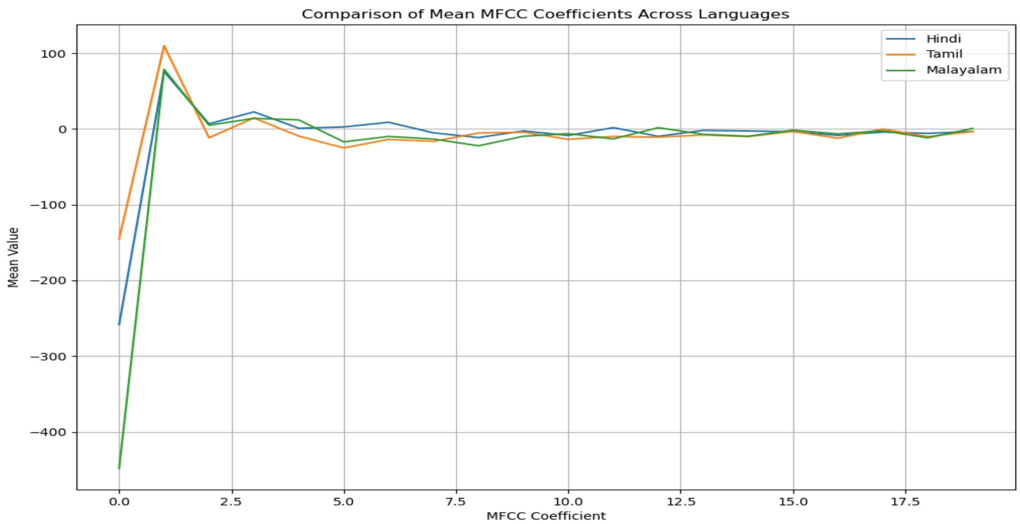
Time



Results and Analysis

MFCC Spectrograms Comparison

The MFCC spectrograms revealed distinctive patterns across the three languages:

- Hindi
  - Shows moderately high energy concentration in the lower MFCC coefficients
  - The first coefficient has a very high negative mean (-258.27) and the highest variance (22378.26)
  - Notable positive mean values in coefficients 2-7, particularly coefficient 2 (75.90)

- Tamil
  - Displays a different pattern with less extreme values in the first coefficient (-145.07)
  - Higher mean value for coefficient 2 (110.11) compared to Hindi and Malayalam
  - More negative mean values across most coefficients, particularly coefficient 6 (-24.96)
  - Generally lower variance values compared to Hindi, especially in the first coefficient (7472.36)

- Malayalam
    - Has the most extreme negative value for the first coefficient (-448.33)
    - Similar pattern to Hindi for coefficient 2 (78.83)
    - Unique positive values for coefficients 5 (11.97) and 20 (0.73)
    - There is a high variance in the first coefficient (22964.08), similar to Hindi

Statistical Analysis

The mean MFCC coefficients comparison reveals:

1. First Coefficient (Energy): Malayalam has the most negative mean value, followed by Hindi and Tamil. This suggests significant differences in overall energy distribution.

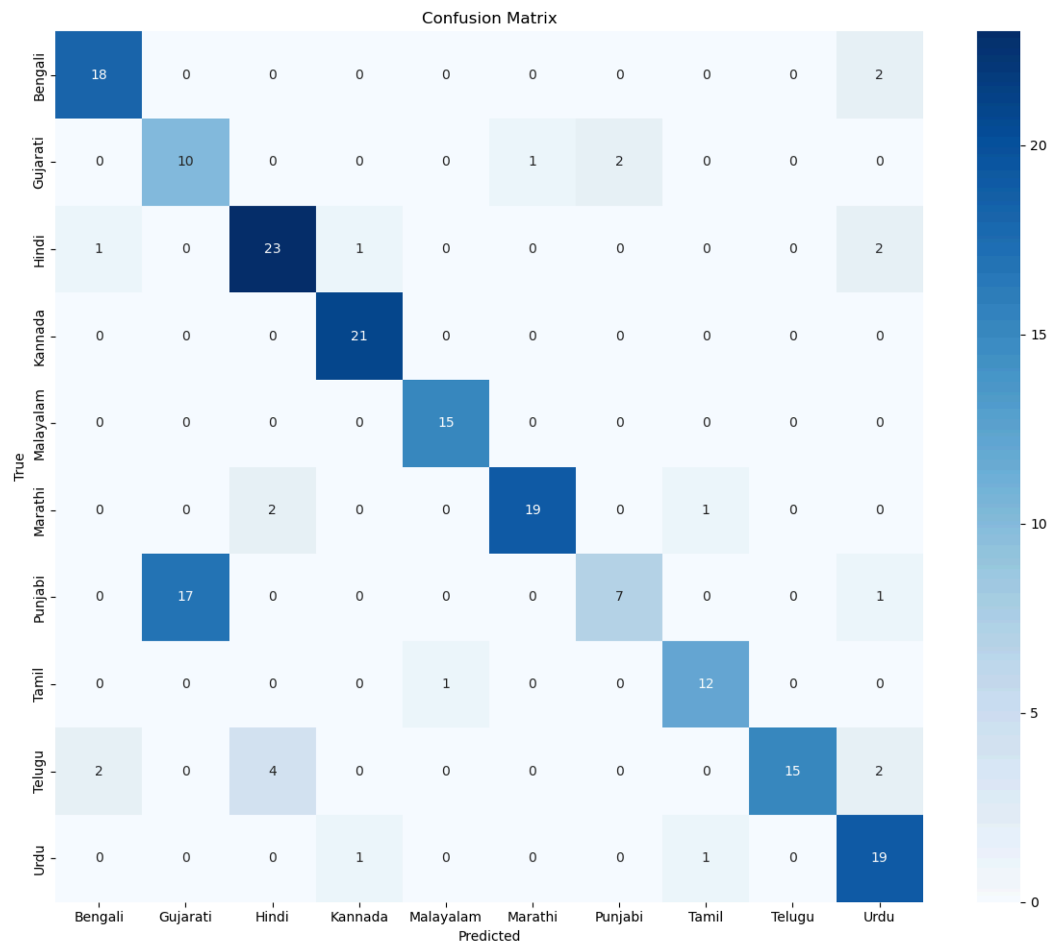
2. Coefficient Patterns:
- Hindi shows moderate variability with both positive and negative means
- Tamil has predominantly negative mean values across most coefficients
- Malayalam shows a mixed pattern with some distinctive positive values

3. Variance Analysis:
- Hindi and Malayalam show similar high variance patterns, particularly in the first few coefficients
- Tamil has comparatively lower variance across most coefficients
- Higher coefficients generally have lower variance across all languages

Task B. Classfication

- Statistical features derived from MFCC coefficients (mean and variance) were used to build a Random Forest classifier
- The dataset was split into training (80%) and testing (20%) sets
- Features were standardized before model training
- Classification performance was evaluated using accuracy and a confusion matrix

Classification Results



The Random Forest classifier achieved effective language discrimination based on MFCC features, demonstrating that the observed differences in spectral characteristics are consistent enough for automated recognition.

Challenges in Using MFCCs for Language Differentiation

1. Speaker Variability: Individual speakers' vocal characteristics can sometimes overshadow language-specific features. Age, gender, and vocal tract differences introduce additional variability.

2. Background Noise: Environmental noise can significantly impact MFCC extraction quality, particularly affecting the higher coefficients, which typically contain finer phonetic details.

3. Regional Accents: Dialectal variations within languages can create overlapping MFCC patterns between different languages spoken with similar regional influences.

4. Content Dependency: The semantic content of utterances can influence the MFCC patterns, potentially creating inconsistencies when comparing different content across languages.

5. Recording Conditions: Variations in recording equipment, distance from the microphone, and room acoustics can introduce artificial differences in MFCC representations.

Refernces

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2. WavLM Chen, S., Wang, C., Chen, Z., Yao, Y., Wu, Y., Chen, J., Zhao, S., Liu, S., & Deng, L. (2022). WavLM: Large-scale self-supervised pre-training for full stack speech processing. arXiv. https://arxiv.org/abs/2110.13900

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